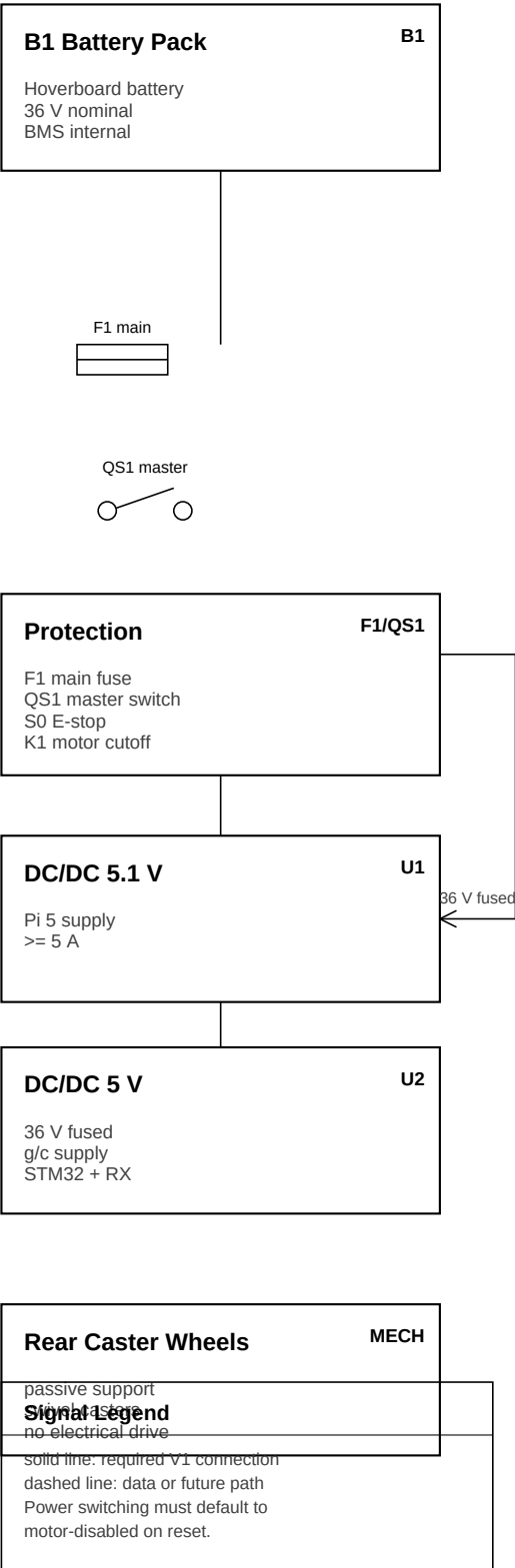


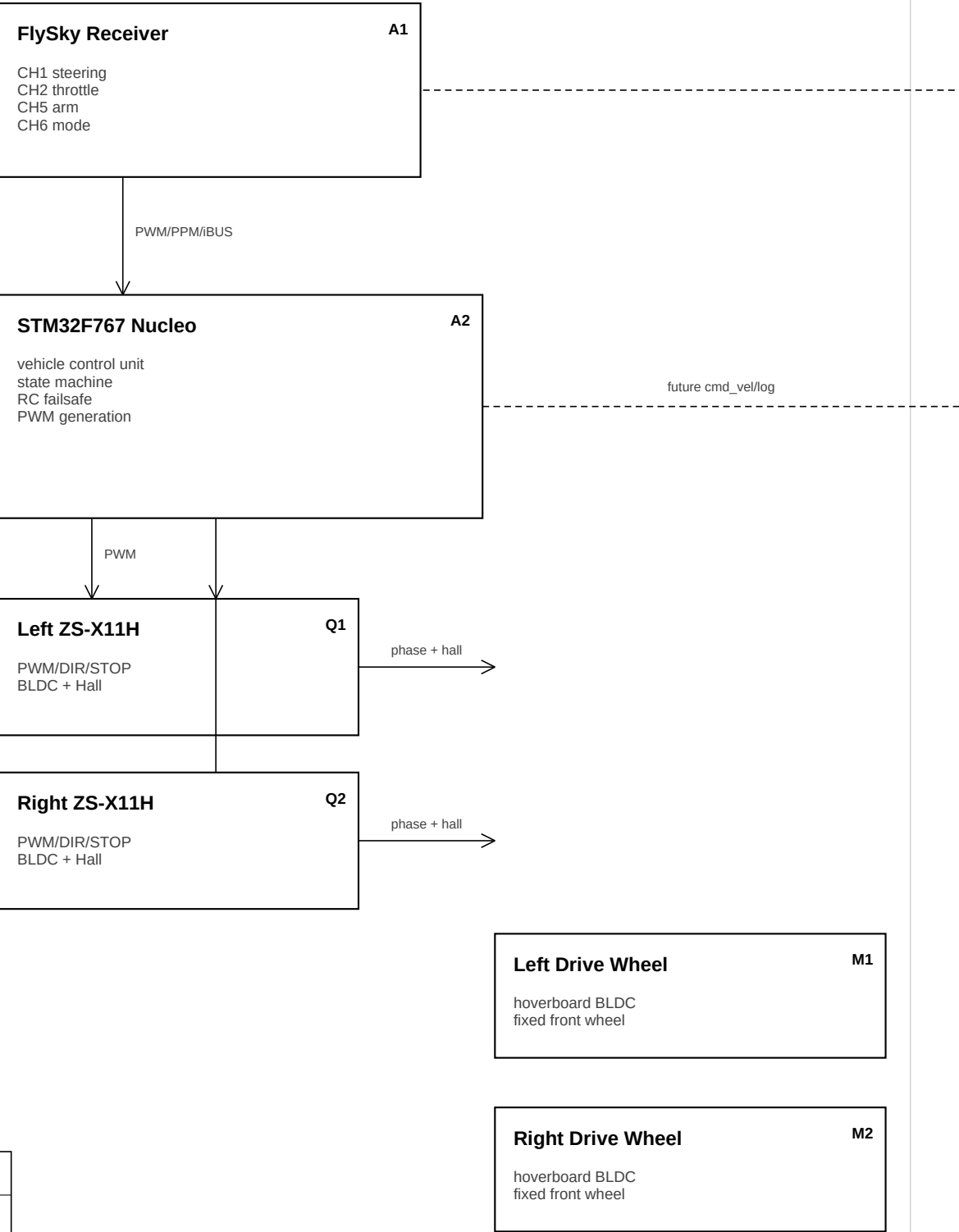
V1 System Architecture

Teleoperation-first electrical concept, no CAN in V1

POWER DOMAIN - 36 V / high current



CONTROL DOMAIN - real-time embedded



PERCEPTION / COMPUTE



Note: Concept-level drawing for prototype development. Verify ratings, pinout, wire gauge and fusing before energizing hardware.

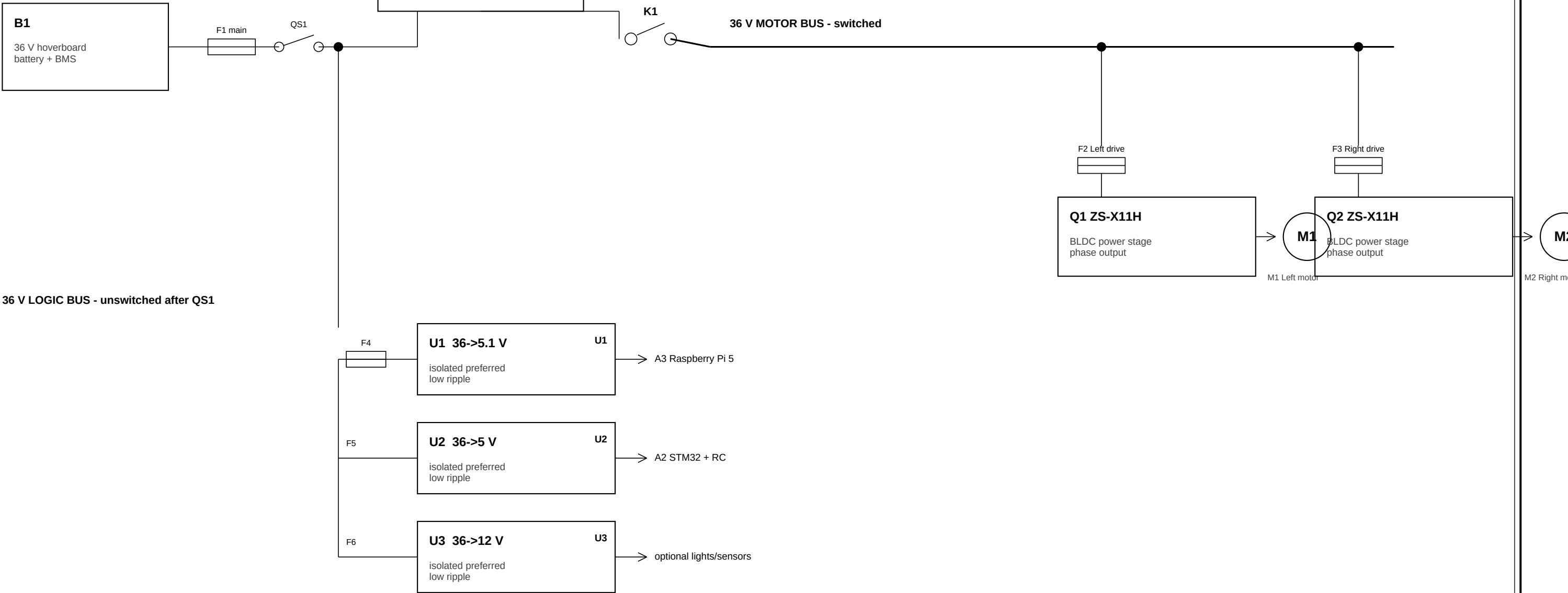
PROJECT	DOCUMENT	SHEET	REV
AgriBot UGV	AGB-EE-DRW-001	01/08	A
TITLE	STATUS		
V1 System Architecture	PRELIMINARY / ENGINEERING CONCEPT		

Power Distribution Single-Line Diagram

Industrial EE Documentation Format

36 V battery system, separated motor and logic branches

Positive distribution



0 V RETURN / CHASSIS BONDING POINT P0 - star connection near battery negative, no motor current through logic ground

Protection Notes

NOTE

F1 installed within 150 mm of battery positive.
F2/F3 protect motor branch wiring.
F4/F5 protect low-voltage converters.
Fuse values shall be selected from measured current and wire gauge.

Recommended Additions

EMC

TVS diode on 36 V bus
bulk capacitor near motor drivers
reverse-polarity protection
pre-charge or soft-start if inrush trips BMS

Reference Designators

IEC

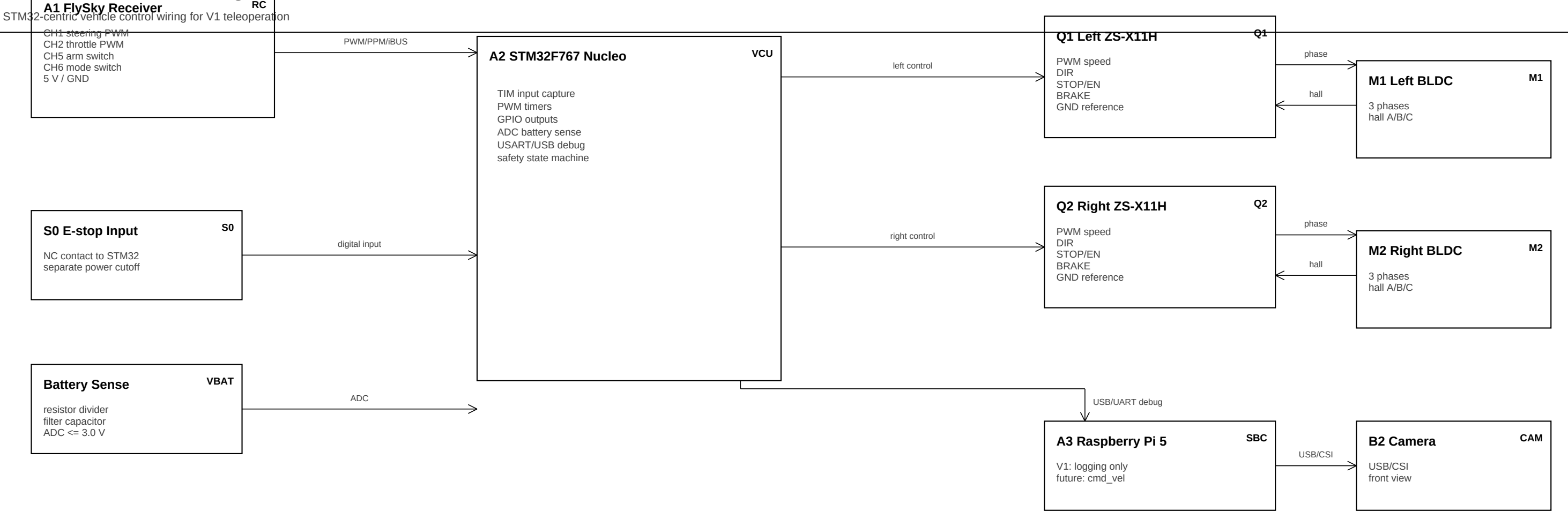
B = battery/sensor
F = fuse
K = relay/contactor
Q = power driver
S = switch
U = converter/module
M = motor
A = assembly/PCB

PROJECT AgriBot UGV	DOCUMENT AGB-EE-DRW-001	SHEET 02/08	REV A
TITLE Power Distribution Single-Line Diagram		STATUS PRELIMINARY / ENGINEERING CONCEPT DATE: 2025-05-23	

Note: Concept-level drawing for prototype development. Verify ratings, pinout, wire gauge and fusing before energizing hardware.

Control I/O and Wiring Overview

Industrial EE Documentation Format



Connector Allocation - prototype harness	HARNESS
J1: 36 V battery and main return J2: left motor driver control and power J3: right motor driver control and power J4: RC receiver and operator controls J5: Raspberry Pi service/debug link J6: E-stop and motor contactor loop	

Electrical Interface Rules	RULES
All STM32 outputs to motor-driver command pins shall fail LOW at reset. Use pull-down resistors on PWM, STOP and BRAKE. Direction changes only permitted after PWM = 0 and a dead time delay.	

Note: Concept-level drawing for prototype development. Verify ratings, pinout, wire gauge and fusing before energizing hardware.

PROJECT	DOCUMENT	SHEET	REV
AgriBot UGV	AGB-EE-DRW-001	03/08	A
TITLE		STATUS	
Control I/O and Wiring Overview		PRELIMINARY / ENGINEERING CONCEPT	

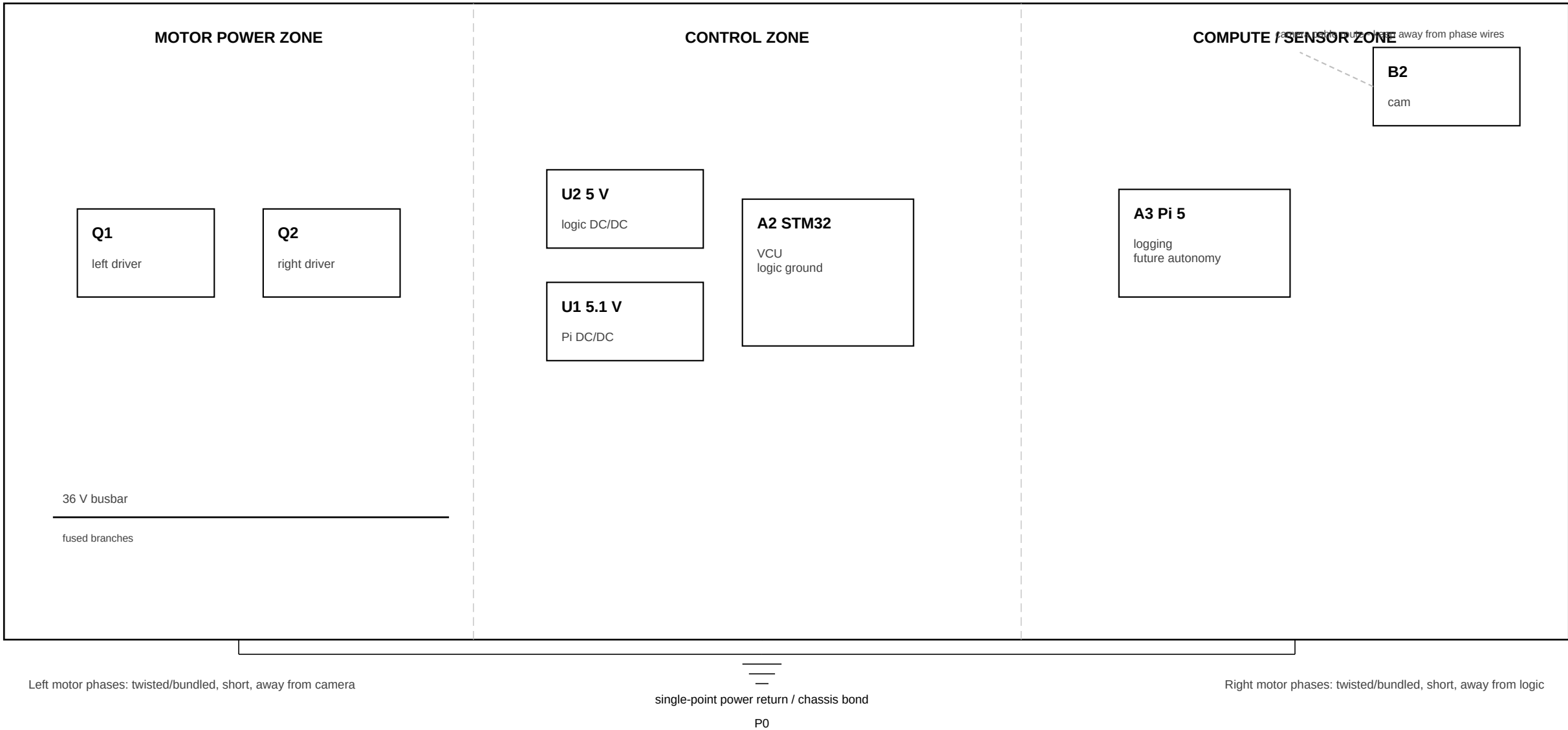
DATE: 2025-05-23

Grounding, EMC and Physical Installation

Industrial EE Documentation Format

Harness routing and electrical zoning for noisy BLDC drive system

Top view electrical zoning - simplified chassis envelope



Grounding Rules	GND
1. Star return at P0. 2. Do not daisy-chain Pi/STM32 ground through motor driver power return. 3. Add ferrites on camera and USB if noise appears.	

Harness Rules	HARNESS
1. Motor phase wires: shortest path. 2. RC/sensor wires: separate bundle. 3. Cross noisy and sensitive cables at 90 degrees.	

Enclosure Rules	MECH
1. Use cable glands and strain relief. 2. Mount electronics low on chassis. 3. Seal against dust, soil and irrigation water.	

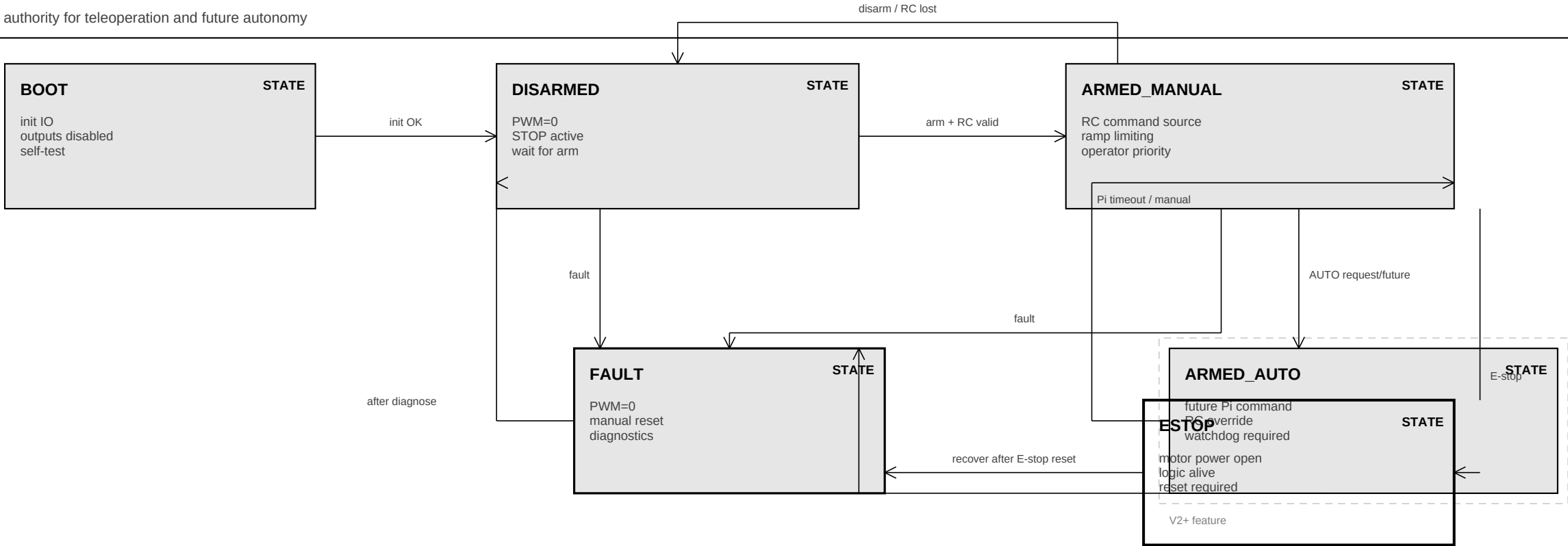
PROJECT	DOCUMENT	SHEET	REV
AgriBot UGV	AGB-EE-DRW-001	04/08	A
TITLE		STATUS	
Grounding, EMC and Physical Installation		PRELIMINARY / ENGINEERING CONCEPT	

Note: Concept-level drawing for prototype development. Verify ratings, pinout, wire gauge and fusing before energizing hardware.

Safety and Operating State Machine

Industrial EE Documentation Format

STM32 state authority for teleoperation and future autonomy



Fault Sources

F

RC loss
battery undervoltage
Pi timeout in AUTO
invalid command
driver/thermal future

Safety Priority Order

1. Hardware E-stop opens motor power path.
2. STM32 fault logic disables PWM/STOP/BRAKE commands.
3. RC manual override has priority over Pi.
4. Future Pi autonomy is accepted only when AUTO mode and watchdog are valid.

Transition Guards

- RC loss in manual mode: stop immediately.
- Pi command timeout in auto mode: command = zero, return to manual or fault according to switch.
- Direction change: ramp PWM to zero, wait, change DIR, ramp up.
- Startup default: motor STOP active and PWM disabled until explicitly armed.

Reset Conditions

R

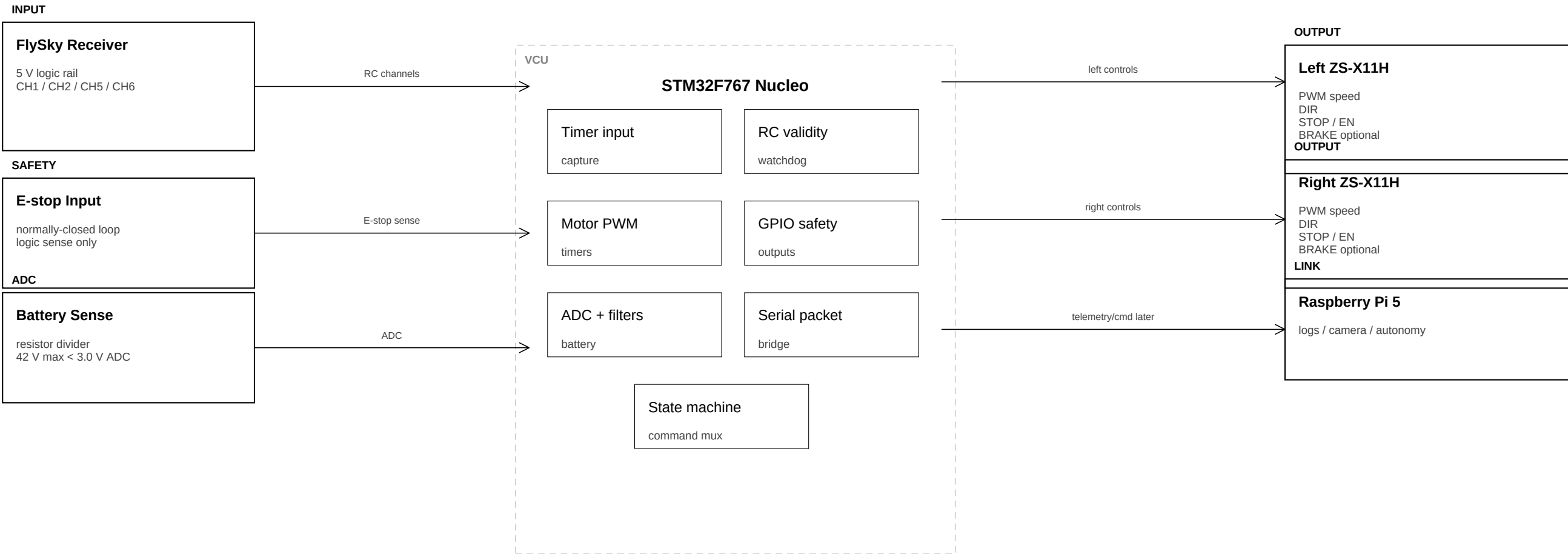
E-stop released
arm switch OFF->ON
fault latch cleared
operator present

Note: Concept-level drawing for prototype development. Verify ratings, pinout, wire gauge and fusing before energizing hardware.

PROJECT	DOCUMENT	SHEET	REV
AgriBot UGV	AGB-EE-DRW-001	05/08	A
TITLE		STATUS	
Safety and Operating State Machine		PRELIMINARY / ENGINEERING CONCEPT	

Interface Control Diagram

Clear electrical interfaces between receiver, STM32, motor drivers, power rails, and companion computer



Interface Notes

- Do not power the Pi from the ZS-X11H 5 V pin.
- The STM32 output default must keep STOP active and PWM at zero.
- Use common logic ground, but do not let motor current return through signal wiring.
- Label every connector with voltage, signal direction, and fuse rating.

PROJECT	DOCUMENT	SHEET	REV
AgriBot UGV	AGB-EE-DRW-001	06/08	A
TITLE	STATUS		
Interface Control Diagram	PRELIMINARY / ENGINEERING CONCEPT		

Serial Communication Protocol

Pi <-> STM32 packet format, message definitions, timing

1. Physical Layer

Interface: USB Serial (Nucleo USB) or dedicated UART
Baud rate: 115200 Data: 8N1 Flow control: none
Cable: USB-A to micro-USB, < 1 m recommended

2. Packet Format

START1	START2	LENGTH	MSG_ID	SEQ	PAYLOAD	CRC16_H	CRC16_L
0xAA	0x55	N	ID	n	N bytes		

CRC-16/CCITT-FALSE over bytes START1..PAYLOAD

3. Messages: Pi -> STM32

MSG_ID	Name	Payload	Rate	Notes
0x01	HEARTBEAT	timestamp (4B uint32)	10 Hz	watchdog keepalive
0x02	CMD_VEL	linear_vel (float32), angular_vel (float32)	20 Hz	V2+ only
0x03	SET_MODE	mode (1B): 0=MANUAL, 1=AUTO	on change	V2+ only

4. Messages: STM32 -> Pi

MSG_ID	Name	Payload	Rate	Notes
0x81	STATUS	state(1B) faults(2B) batt_mv(2B) rc_ok(1B)	10 Hz	always sent
0x82	MOTOR_FB	left_duty(i16) right_duty(i16) left_dir(1B) right_dir(1B)	20 Hz	when armed
0x83	RC_CHANNELS	ch1..ch6 (uint16 x 6, little-endian)	20 Hz	always sent

5. Safety Rules

- STM32 stops motors if no valid HEARTBEAT from Pi for 250 ms.
- CRC mismatch: discard packet silently, increment error counter.
- RC input always has priority in MANUAL mode.
- STM32 never accepts CMD_VEL unless in ARMED_AUTO state.
- Pi should monitor STATUS messages; if absent for 500 ms, alert operator.
- Sequence number wraps 0-255; receiver uses it for packet loss detection.

PROJECT	DOCUMENT	SHEET	REV
AgriBot UGV	AGB-EE-DRW-001	07/08	A
TITLE		STATUS	
Serial Communication Protocol		PRELIMINARY / ENGINEERING CONCEPT	

STM32 Pin Allocation and Wiring Table

Preliminary assignment - finalize after CubeMX configuration

Function	STM32 Pin	Periph	Signal	Direction	Destination	Notes
Left PWM	PA8 (TBD)	TIM1_CH1	PWM 2kHz	OUT	Q1 ZS-X11H speed	0-100% duty
Left DIR	PB0 (TBD)	GPIO	Digital	OUT	Q1 ZS-X11H DIR	level TBD
Left STOP	PB1 (TBD)	GPIO	Digital	OUT	Q1 ZS-X11H STOP	default=disabled
Left BRAKE	PB2 (TBD)	GPIO	Digital	OUT	Q1 ZS-X11H BRAKE	optional
Right PWM	PA9 (TBD)	TIM1_CH2	PWM 2kHz	OUT	Q2 ZS-X11H speed	0-100% duty
Right DIR	PC0 (TBD)	GPIO	Digital	OUT	Q2 ZS-X11H DIR	level TBD
Right STOP	PC1 (TBD)	GPIO	Digital	OUT	Q2 ZS-X11H STOP	default=disabled
Right BRAKE	PC2 (TBD)	GPIO	Digital	OUT	Q2 ZS-X11H BRAKE	optional
RC iBUS	PD6 (TBD)	USART2_RX	iBUS 115200	IN	A1 FlySky RX	single wire all CH
RC CH1 alt	PA0 (TBD)	TIM2_CH1	PWM capture	IN	A1 CH1 steering	if PWM mode
RC CH2 alt	PA1 (TBD)	TIM2_CH2	PWM capture	IN	A1 CH2 throttle	if PWM mode
RC CH5 alt	PA2 (TBD)	TIM2_CH3	PWM capture	IN	A1 CH5 arm	if PWM mode
E-stop sense	PE0 (TBD)	GPIO	Digital	IN	S0 NC contact	pull-up, active LOW
Battery ADC	PA3 (TBD)	ADC1_IN3	Analog	IN	VBAT divider	120k/10k, cap filter
Pi UART TX	PD5 (TBD)	USART2_TX	Serial 115200	OUT	A3 Pi 5	telemetry
Pi UART RX	PD6 (TBD)	USART2_RX	Serial 115200	IN	A3 Pi 5	cmd_vel (V2+)
USB debug	USB	USB OTG	Serial	BIDIR	PC / Pi	Nucleo USB
Status LED	PB7 (TBD)	GPIO	Digital	OUT	LED	onboard or external
Buzzer	PB8 (TBD)	TIM/GPIO	PWM/Digital	OUT	Piezo buzzer	fault/arm indication

Pin Assignment Notes

All pins marked (TBD) - finalize in STM32CubeMX.
Prefer 5V-tolerant pins for ZS-X11H interface.
Use internal pull-downs on PWM/STOP/BRAKE outputs.
Battery ADC: verify divider ratio against actual battery voltage range before first power-on.
iBUS and Pi UART may share USART2 - allocate separate UARTs if both are used simultaneously.

Wire Gauge Reference

Motor phase wires: use original hoverboard gauge
Motor control signals: 24-26 AWG
Power to Pi: 18 AWG minimum
Power to STM32/logic: 22 AWG
RC receiver: 24-26 AWG
E-stop loop: 22 AWG

PROJECT	DOCUMENT	SHEET	REV
AgriBot UGV	AGB-EE-DRW-001	08/08	A
TITLE		STATUS	
STM32 Pin Allocation and Wiring Table		PRELIMINARY / ENGINEERING CONCEPT	